matheus hoffmann

the aquatic prometheus

sr. software engineer and jr. researcher, with 8 years of experience building scalable solutions, mainly for payment processing and fraud-fighting. my research topics are wave modelling, game theory and self-improving apps. i work in the open, and i'm open to commercial offers.

mhoffmannfs@gmail.com · linkedin · github · the site · ways of working

the arc

backend, then mlops, then ml, then back to the sea: research and foss.

2026	**marola** and **Ouroboric**: open research, with friends.
2023–2024	**Signifyd** (US): senior software and ml engineer, fraud-fighting.
2021–2022	**Elemeno AI**: mlops engineer.
2020	**Broad** (YC W21): senior software engineer, a digital bank.
2017–2019	**Stone**: software engineer, payments.
2016–2017	**LIOC**: the ocean instrumentation lab of UFRJ.

also TXP (UK, formerly Vigil). the whole story, with the knowledge graph of the works, is on the site.

the works, each with a DOI

marola 10.5281/zenodo.23224155
a citizen-science ocean intelligence layer for the brazilian coast: "can i swim tomorrow, and when?". scala 3, rag with citations, designed in public (MIPs).

ww3-gpu 10.5281/zenodo.23221351
a kernel of NOAA's WAVEWATCH III wave model, ported to c++ and kokkos. it matches the fortran bit for bit on serial, openmp and cuda.

marola-bench 10.5281/zenodo.23352633
a proposal: a WeatherBench for brazil, scored against buoys.

picasso 10.5281/zenodo.23311208
my nix labs: flakes that pin every toolchain the other works build with.

closed source: the god that failed
10.5281/zenodo.23333656
a position paper on reverse engineering and the ethics of citation in the age of coding agents. not peer reviewed.

self-improving apps
writing and doing research for ouroboric-dev: open factories that improve themselves.

how i work

async and remote first, with 4h of overlap a day at most. no live coding: a code interview, or a talk about past projects. money first. references on request.

certs

AWS certified 4x: Cloud Practitioner, Machine Learning Specialty, Machine Learning Engineer Associate, Data Engineer Associate. **PADI certified 3x**: Advanced Open Water Diver, Enriched Air Nitrox Diver, Rescue Diver.

the sea

3rd place, Oceanman 10 km, rio de janeiro, september 2025. 2× the 10 km. strava.